

BMEG3105 Data Analytics for Personalized Genomics and Precision Medicine

Lecture 8: Classification 02/10/2026

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8.0 Outline:

1. Quick Recap
2. Classification
3. K-Nearest Neighbors
4. Clustering vs. Classification
5. Unsupervised vs Supervised Learning

8.1 Quick Recap:

Mahalanobis Distance:

- Calculating distance considering the data distribution

Hierarchical Clustering:

- Produces a set of nested clusters organized as a hierarchical tree (tree diagram)
 - Dendrogram
- Steps:
 - Compute similarity/distance matrix
 - Let each data point be a cluster
 - Merge the two closest clusters + update the matrix
 - Repeat until a single cluster remains

Clustering:

- Using similarity or dissimilarity to measure the differences
- Intra-cluster distances are small
- Inter-cluster distances are large
- Distance measured by:
 - Cosine similarity
 - Correlation
 - Euclidean distance
 - Manhattan distance
 - Mahalanobis distance

8.2 Classification

- **Why?**
 - Find characteristics of each class
 - Classifying items
 - For better organization
 - Also helps us know where to put the new items
 - Classifying people
 - For patients: to find different treatments for different groups (like by age sometimes)
 - For customers: to see if a person is within the target group
 - In biology: given a new gene expression profile, is it normal or a tumor?
- **What is classification**
 - Given a collection of records → **the training set**
 - Each record contains a set of **attributes**
 - One of which is the **class**
 - We just need a method to assign the class for some unseen records based on the other information (attributes) and the training set → needs to be as accurate as possible
- **How?**
 1. Have training data with a class (that we want to classify)
 2. Need a classification method
 3. Have data to be classified
 4. Classify!

8.3 Classification Methods

K-nearest neighbor (KNN)

- Simple algorithm that stores all available instances + classifies new instances based on a **distance metric** to the available ones

Overall idea → if it matches the characteristics of a dog (walk, bark, ...), then it's probably a dog

- Training Process: store the available training instances
- Predicting Process:
 - Find the **K** training instances that are closest to the query instance
 - Return the most frequent class label among those K instances
- **Data needs to be normalized**

What do we need to determine when using KNN?

- **Distance metric** → how do we decide how close they are?

- Cosine similarity
- Correlation
- Euclidean, Manhattan, or Mahalanobis distance
- **How many neighbors: K**
 - Usually in practice, using K between 5 to 10 gives good results for most low-dimensional data sets
 - Good K can also be chosen by using cross-validation (explained later in Lecture 11)
- **Weighing function (optional)**

Standard Procedure of KNN

Once we have chosen the distance metric and K:

1. Normalization
2. Compute Distances
3. Identify the K most similar data
4. Take their class out and find the mode class

For example:

Person	Height(m)	Weight(kg)	Gender
P1	1.79	75	M
P2	1.64	54	F
P3	1.70	63	M
P4	1.88	78	M
P5	1.75	70	??

We have chosen the **Euclidean distance** and **K = 2**

1. Normalization

Person	Height	Weight	Gender
P1	0.625	0.875	M
P2	0	0	F
P3	0.25	0.375	M
P4	1	1	M
P5	0.4583	0.6667	??

a.

b. Using: $x_{norm} = \frac{x-min}{max-min} \rightarrow$ so min is P2, max is P4 values

c. For example: P1 weight: $\frac{1.79-1.64}{1.88-1.64} = 0.625$

2. Compute distances

Person	P5	Gender
P1	0.267	M
P2	0.809	F
P3	0.358	M
P4	0.636	M
P5	0	??

a.

b. Where: $distance = \sqrt{(height_{P_5} - height_{P_i})^2 - (weight_{P_5} - weight_{P_i})^2}$

3. Identify the K most similar data

Person	P5	Gender
P1	0.267	M
P2	0.809	F
P3	0.358	M
P4	0.636	M
P5	0	??

a.

b. Closest to 0 in this

4. Take their class out + find the mode class

a. Since the other two (most similar) are male, we can classify P5 as **M**

Recap:

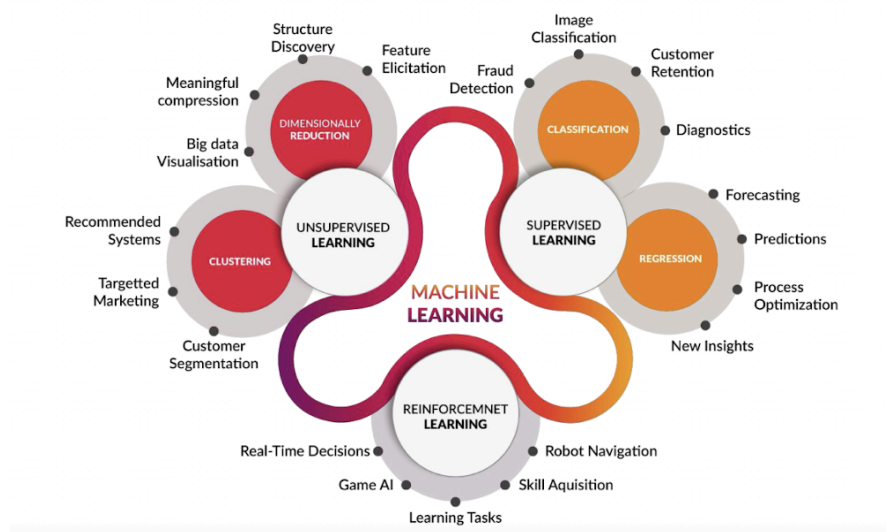
8.4 Clustering vs Classification

Clustering	Classification
Similarity measurement → clustering algorithm	Classification method → training data with class
Want to find similarity/clusters in the data	Want to assign the class to new data
Unknown number of classes	Known number of classes
Outputs the cluster index for each point	Outputs the class assignment of testing data
One-phase algorithm	Two-phase algorithm (training + application)

8.5 Unsupervised learning vs supervised learning

Unsupervised Learning	Supervised Learning
Machine Learning algorithms to analyze and cluster unlabelled data	Machine Learning algorithms to classify and predict outcomes; trained on labelled data
Ex: clustering + dimension reduction	Ex: classification + regression

Machine Learning:



Using KNN in Python

<https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>

Resources + Uncovered Topics

- Introduction to data mining: Chapter 4.1, 4.2 & 5.2
- Problem of KNN (next lecture)
- Logistic regression and neural networks (Next lecture)
- Decision tree/SVM/Bayesian
- Model overfitting
- Cross-validation (Lecture 9)